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Cutting Assembly For Surgical Instrument With Clog-Reducing Tip

Abstract

A cutting assembly configured to be removably coupled to a drive assembly of a surgical instrument. An inner tube is coaxially and rotatably disposed within an outer tube and defines a lumen in communication with a cutting window. A projection is disposed within the inner tube. At least a portion of the projection is positioned distal to a proximal boundary of the cutting window. The projection may define a throat having a distance between the proximal boundary of the cutting window on the outer tube and a nearest point on the projection. The throat may be narrower than a diameter of the lumen. An inner surface of the projection permits cutting action to continue until the material is sufficiently reduced to pass through the throat. The projection may be formed by an insert bonded within the inner tube.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of copending U.S. patent application Ser. No. 18/227,405, filed on Jul. 28, 2023, which is a continuation of U.S. patent application Ser. No. 17/331,795, filed on May 27, 2021, now U.S. Pat. No. 11,766,274, which is a continuation of U.S. patent application Ser. No. 16/316,500, filed on Jan. 9, 2019, now U.S. Pat. No. 11,020,139, which is United States national entry of International Patent Application No. PCT/US2017/042101, filed on Jul. 14, 2017, which claims priority to and all the benefits of U.S. Provisional Patent Application No. 62/362,117, filed on Jul. 14, 2016, the entire contents of each are hereby incorporated by reference.

TECHNICAL FIELD
[0002] The present disclosure relates generally to surgical instruments and, more particularly to, a surgical instrument with a clog reducing tip for use on patients.
BACKGROUND
[0003] It is known that medical practitioners have found it useful to use surgical instruments to assist in the performance of surgical procedures. A surgical instrument is designed to be applied to a surgical site on the patient. The practitioner is able to position the surgical instrument at the site on the patient at which the instrument is to perform a medical or surgical procedure. Endoscopic surgical procedures are routinely performed in order to accomplish various surgical tasks. In an endoscopic surgical procedure, small incisions, called portals, are made in the patient. An endoscope, which is a device that allows medical personnel to view the surgical site, is inserted in one of the portals. Surgical instruments used to perform specific surgical tasks are inserted into other portals. The surgeon views the surgical site through the endoscope to determine how to manipulate the surgical instruments in order to accomplish the surgical procedure. An advantage of performing endoscopic surgery is that, since the portions of the body that are cut open are minimized, the portions of the body that need to heal after surgery are likewise reduced. Moreover, during an endoscopic surgical procedure, only relatively small portions of the patient's internal organs and tissue are exposed to the open environment. This minimal opening of the patient's body lessens the extent to which a patient's organs and tissue are open to infection.
[0004] Many tube devices have been developed for use in surgical procedures. They are valuable because they facilitate reduced incision size, improved access and visibility, while enhancing surgical outcome and quicker recovery. Some are cutting devices having either two tubes, one within another, or a single tube with a cutting window. Such cutting devices may be an ear, nose, and throat (ENT) shaver devices.
[0005] Clogging of ENT shaver devices is a common annoyance during endoscopic sinus surgery. A common cause of clogging is the trapping of sinus bone and tissue at the distal tip of the ENT shaver device just proximal of a cutting window. Another common cause of clogging is the trapping of sinus bone and tissue just proximal the tube(s) of the cutting device.
[0006] A surgical instrument that overcomes these challenges is desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Advantages of the present disclosure will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

[0008] FIG. **1** is a perspective view of a surgical instrument according to an exemplary embodiment of the present disclosure.

[0009] FIG. **2** is an exploded perspective view of the surgical instrument of FIG. **1** with a drive assembly removed.

[0010] FIG. **3** is a perspective view of a clog-reducing tip of a cutting assembly of the surgical instrument of FIGS. **1** and **2** in accordance with an exemplary embodiment of the present disclosure.

[0011] FIG. **4** is a fragmentary elevational view of the clog-reducing tip of the cutting assembly of FIG. **3** with removed material represented schematically.

[0012] FIG. **5** is another fragmentary elevational view of the clog-reducing tip of the cutting assembly of FIG. **3**.

[0013] FIG. **6** is another fragmentary elevational view of the clog-reducing tip of the cutting assembly of FIG. **3**.

[0014] FIG. **7** a cross-sectional view of the cutting assembly of FIG. **3** taken along lines **7-7**.

[0015] FIG. **8** is a perspective view of the clog-reducing tip according to another exemplary embodiment of the present disclosure.

[0016] FIG. **9** is fragmentary perspective view of the clog-reducing tip of FIG. **8**.

[0017] FIG. **10** is a perspective view of the clog-reducing tip according to another exemplary embodiment of the present disclosure.

[0018] FIG. **11** is fragmentary perspective view of the clog-reducing tip of FIG. **10**.

[0019] FIG. **12** is a perspective view of the clog-reducing tip according to another exemplary embodiment of the present disclosure.

[0020] FIG. **13** is elevational view of the clog-reducing tip of FIG. **12**.

[0021] FIG. **14** is an angled view of the clog-reducing tip of FIG. **12**.

[0022] FIG. **15** is a plan view of the clog-reducing tip of FIG. **12**.

[0023] FIGS. **16-19** are diagrammatic views illustrating a machined process for forming the clog-reducing tip of FIGS. **12-15**.

[0024] FIG. **20** is a cross-sectional view of the surgical instrument of FIG. **1** taken along lines **20-20**.

DETAILED DESCRIPTION

[0025] Referring to FIG. **1**, one embodiment of a surgical instrument **10**, according to the present disclosure, is shown for use in a medical procedure for a patient (not shown). In one embodiment, the surgical instrument **10** is an ENT shaver that is disposable and used for resecting sinus bone and tissue during endoscopic sinus surgery. As illustrated, the surgical instrument **10** includes a drive assembly, generally indicated at **12** and shown in phantom lines, and a cutting assembly, generally indicated at **14**, removably coupled to the drive assembly **12**. The drive assembly **12** is used to rotate a portion of the cutting assembly **14** to remove tissue, bone, etc. from a surgical site of the patient. It should be appreciated that the surgical instrument **10** may be operated by a user (not shown) such as a surgeon.

[0026] As illustrated in FIG. **1**, the drive assembly **12** includes a housing **15** extending axially. The housing **15** is generally cylindrical in shape. The drive assembly **12** also includes a motor **16** disposed in the housing **15** and having a rotatable drive element **18** coupled to the cutting assembly **14**. The motor **16** may be of an electric or pneumatic type. In one embodiment, the drive element

18 is removably coupled to the cutting assembly **14**.

[0027] It should be appreciated that, in one embodiment, the cutting assembly **14** may be free of any motor. Thus, the cutting assembly **14** may be configured to be disposable after a single-use, or series of uses. Because the cutting assembly **14** may not include any motors, the cost of the cutting assembly **14** may be reduced.

[0028] Referring to FIGS. **1-7**, the cutting assembly **14** includes a plurality of tubes or tube assembly, generally indicated at **20**, extending axially between a distal end **23** and a proximal end **21** (FIG. **20**) opposite the distal end **23**. The tube assembly **20** has a longitudinal axis **24** defined between the proximal end **21** and the distal end **23**. The tube assembly **20** includes a window **22**, for example, a cutting window, near or at the distal end **23** with the cutting window **22** adapted to be applied to a surgical site of a patient. In certain embodiments, the tube assembly **20** includes a first, or outer, tube **26** and a second, or inner, tube **28**. The inner tube **28** is coupled to the drive assembly **102** and rotatable by the drive element **18** relative to the outer tube **26**. The inner tube **28** may be removably coupled to the drive element **18**, for example, in an embodiment where the cutting assembly **14** is disposable after a single-use or series of uses.

[0029] In one embodiment, the outer tube **26** is non-rotatable and the inner tube **28** is rotatable relative to the outer tube **26**. The inner tube **28** and has a lumen **30** extending between the proximal end **21** and the distal end **23** of the tube assembly **20**. The inner tube **28** may comprise a proximal region **32** and a distal region **34** to be described. The inner tube **28** comprises, forms, or defines a first or inner cutting window **36** at or near the distal end **23** of the tube assembly **20**, such as within the distal region **34** of the inner tube **28**.

[0030] Each of the inner tube **28** and the outer tube **26** may be generally hollow cylinders extending axially and have a generally circular cross-sectional shape. The outer tube **26** has a diameter greater than a diameter of the inner tube **28** such that the inner tube **28** is disposed within the outer tube **26**. In other words, the outer tube **26** has a lumen extending between the proximal end **21** and the distal end **23** of the tube assembly **20** with the inner tube **28** at least partially disposed within the lumen of the outer tube **26**. In one embodiment to be described (see FIG. **20**), the inner tube **28** has an axial length longer than an axial length of the outer tube **26** such that the inner tube **28** extends past a proximal region **38** of the outer tube **26** when the inner tube **28** is disposed within the outer tube **26**.

[0031] The outer tube **26** may comprise the proximal region **38** and a distal region **40** as shown in FIG. **2**. The outer tube **26** forms a second or outer cutting window **42** at or near the distal end **23** of the tube assembly, such as within the distal region **40** of the outer tube **26**. The inner cutting window **36** and the outer cutting window **42** define the cutting window **22** of the tube assembly **20**. In one exemplary embodiment, the outer tube **26** may include a radial reduction step **44** within the distal region **34** to allow an outer surface of the inner tube **28** and an inner surface of the outer tube **26** to be close together.

[0032] In one embodiment, the tube assembly **20** may further include a non-rotatable sheath or third or covering tube **46** disposed over a portion of the outer tube **26**. The covering tube **46** has an axial length less than an axial length of the outer tube **26**. The covering tube **46** may be angled, straight, or malleable. It should be appreciated that the covering tube **46** is optional. In addition, it should be appreciated that the covering tube **46** is coupled to a connecting hub **68** to be described. Furthermore, it should be appreciated that any suitable tubing configuration may be utilized so long as the cutting assembly **14** defines the cutting window **22** and can be driven by the drive assembly **12**.

[0033] The inner tube **28** and outer tube **26** are made of a metal material such as stainless steel or a non-metallic material such as a composite depending on the application. The covering tube **46** may be made of a metal material or a non-metallic material such as a composite depending on the application. It should be appreciated that a wall thickness of the inner tube **28** and the outer tube **26** is relatively thin such as approximately 0.1 to approximately 0.5 millimeters (mm) to allow the tube assembly **20** to be of a relatively small diameter and also to be lightweight. It should also be appreciated that the diameters of the inner tube **28** and the outer tube **26** have a relatively small diameter such as approximately 2.0 mm to approximately 5.0 mm so as to work in a small opening of a nasal cavity or oral cavity of the patient and to prevent the user's view from being obstructed. In one embodiment, the tube assembly **20** may have a bend (not shown) near the distal end **23**. It should further be appreciated that the inner tube **28** and the outer tube **26** may be scaled larger or smaller depending on the application.

[0034] The cutting assembly **14** also includes a drive hub, generally indicated at **48**, disposed about a proximal end of the inner tube **28** to allow the inner tube **28** to be connected to the drive element **18** for rotation of the inner tube **28** about the longitudinal axis **24**. The drive hub **48** includes a hub member **50** disposed about the inner tube **28**. The hub member **50** extends axially and is generally cylindrical in shape. The hub member **50** has an aperture **52** extending axially at least partially therethrough to receive the inner tube **28** as illustrated in FIG. **2**. The hub member **50** may also include a plurality of ridges **54** extending radially and axially and spaced circumferentially thereabout. The hub member **50** may further include a reduced diameter portion **56** adjacent the ridges **54**. The reduced diameter portion **56** of the hub member **50** defines a reduced aperture **53** in communication with the aperture **52** with the reduced aperture **53** being smaller in diameter than the aperture **52** (see FIG. **20**). The decrease in diameter from the aperture **52** to the reduced aperture **53** forms a lip **55** adapted to be positioned adjacent to or in an abutting relationship with the proximal end **21** of the tube assembly **20** in a manner to be described. The hub member **50** also includes a flange **58** extending radially at a distal end thereof. The hub member **50** may be made of a non-metallic material. The hub member **50** may be integral, unitary, and formed as one-piece.

[0035] The drive hub **48** can also include a spring **60** and a seal **62** such as an o-ring disposed about the hub member **50** at a proximal end thereof in the reduced diameter portion **56**. The drive hub **48** may include a washer **64** and a seal **66** such as an o-ring at a distal end thereof disposed about the distal end of the inner tube **28**. It should be appreciated that the drive hub **48** allows for rotation of the inner tube **28** and may allow for the transfer of fluid through the inner tube **28**. It should also be appreciated that a variety of drive coupling configurations may be used with the cutting assembly **14**.

[0036] The cutting assembly **14** further includes a connecting hub, generally indicated at **68**, disposed about the inner tube **28** and a portion of the drive hub **48** to allow the drive assembly **12** to be removably coupled to the cutting assembly **14**. The connecting hub **68** includes a housing hub **70** adapted to be engaged by a least a portion of a hand of a user and supporting the outer tube **26** or the covering tube **46**. The housing hub **70** includes an aperture **72** extending axially therethrough to receive the outer tube **26** or the covering tube **46**. The housing hub **70** may include a plurality of grip members **74** extending radially and axially and a flange **76** extending radially outwardly at one end to support one or more fingers of a hand. The connecting hub **68** also includes a coupling member **78** disposed about the inner tube **28**. The coupling member **78** extends axially and is generally cylindrical in shape. The coupling member **70** has an aperture **72** extending axially therethrough to receive the inner tube **28**. The coupling member **78** includes a cavity **80** extending axially into the proximal end thereof to receive a distal end of the fluid coupling. The coupling member **78** may include one or more ridges **82** extending radially and spaced circumferentially from each other at the proximal end to be coupled to the housing **15** of the drive assembly **12**. The coupling member **78** may include one or more grooves **84** extending radially inward and circumferentially and spaced axially from each other and one or more seals **86** such as o-rings disposed in the grooves **84**. The connecting hub **68** is made of a non-metallic material. The connecting hub **68** may be integral, unitary, and formed as one-piece. It should be appreciated that the connecting hub **68** allows for the coupling of the drive assembly **12** to the cutting assembly **14**.

[0037] Referring to FIG. **3**, the cutting window **22** includes the inner cutting window **36** in the inner tube **28** formed as an opening extending axially and diametrically through a wall on one side near the distal end **23** of the tube assembly **20**. The cutting window **22** also includes the outer cutting window **42** in the outer tube **26** formed as an opening extending axially and diametrically through a wall on one side near the distal end **23** of the tube assembly **20**. The inner and outer cutting windows **36** and **42** are generally elongated and oval in shape, but may be any suitable shape. The inner cutting window **36** may include at least one or more cutting edges **90**. The cutting edge **90** may include a plurality of teeth **92** forming a serrated edge. The outer cutting window **42** may include at least one or more cutting edges **94**. The cutting edge **94** may include a plurality of teeth **92** forming a serrated edge. The inner cutting window **36** is adapted to be temporarily aligned radially with the outer cutting window **42** to receive material within the cutting window **22** as the inner tube **28** rotates within the outer tube **26**. As the inner tube **28** rotates within the outer tube **26**, the inner and outer cutting windows **36** and **42** are removed from radial alignment such that the cutting edges **90** and **94** cut or reduce the material positioned within the cutting window **22** of the tube assembly **20**.

[0038] In one embodiment illustrated in FIG. **1**, the surgical instrument **10** includes an irrigation connection **95** on the housing **15** for connection to a fluid source and an irrigation path or passage **96** extending through the housing **15** between the irrigation connection **95** and the cutting assembly **14** and between the inner tube **28** and the outer tube **26** to the cutting window **22** to provide lubrication. The surgical instrument **10** also includes an aspiration or suction connection **97** on the housing **15** for connection to a suction source and an aspiration or suction path or passage **98** extending through the housing **15** between the suction connection **97** and the first cutting window **36** of the inner tube **28**.

[0039] FIGS. **4-6** show fragmentary elevational views of a clog-reducing tip, generally indicated at **104**, in accordance with an exemplary embodiment of the present disclosure. Referring first to FIG. **4**, the cutting window **22** of the tube assembly **20** comprises a distal boundary **103** and a proximal boundary **101** opposite the distal boundary **103**. In one embodiment, the boundaries **101** and **103** may be defined as an imaginary plane extending perpendicularly to the longitudinal axis **24** of the tube assembly **20** at a proximal-most point and a distal-most point of the cutting window **22**, respectively. Thus, in the exemplary embodiment illustrated in FIG. **4**, the outer tube **26** projects distal to the inner tube **28** to define the proximal boundary **101** of the cutting window **22**, and the distal end **23** the inner tube **28** is positioned proximally (i.e., within) the outer tube **26** to define the distal boundary **103** of the cutting window **22**. Stated differently, the proximal and distal boundaries **101** and **103** may be considered the proximal-most and distal-most points of the cutting window **22**, respectively, when the cutting window **22** is viewed in plan. In certain embodiments, a portion of the inner tube **28** distal to the proximal boundary **101** of the cutting window **22** defines the distal region **34** of the inner tube **28**.

[0040] As illustrated in FIG. **4**, the clog-reducing tip **104** of the tube assembly **20** comprises a projection **112** within the lumen **30** of the inner tube **28**. The projection **112** is adapted to reduce the size of material removable through the cutting window **22**, thereby reducing clogging of the tube assembly **20**. In certain embodiments, at least a portion of the projection **112** is positioned distal to the proximal boundary **101** (in a direction of arrow **102** in FIG. **4**) to provide a reduced cross sectional area to said lumen **30** relative to the cross sectional area of the lumen **30** proximal to the projection **112**. For another example, the projection **112** occupies a volume V.sub.112 (FIG. **5**) within the distal region **34** of the inner tube **28**. The projection **112** reduces the amount by which material **106** to be removed may penetrate the cutting window **22**. Consequently, the cutting action from rotating the inner tube **28** within the outer tube **26** (via the cutting edges **90** and **94**) reduces the material **106** into sufficiently small bits before the material **106** may pass within the lumen **30** proximal to the cutting window **22**, thereby decreasing the likelihood of clogging of the tube assembly **20**.

[0041] In certain embodiments, the reduced cross sectional area of the lumen **30** may defined as the difference between the cross sectional area of the lumen **30** (e.g., x*d with d being the diameter of the lumen **30**) and a cross sectional area of the projection **112**. In one example, a ratio of the reduced cross sectional area of the lumen **30** to the cross sectional area of the lumen **30** is within the range of 1:1.1 to 1:2.0, and more particularly within the range 1:1.3 to 1:1.8, and even more particularly within the range of 1:1.5 to 1:1.6. The reduced cross sectional is adapted to ensure that no dimension of material **106** (e.g., bone and/or tissue chip) is larger than the cross sectional area of the lumen **30**, and more particularly less than the cross sectional area of the lumen **30** by a predetermined factor based on the ratio described above. In other exemplary embodiments, the volume V.sub.112 of the projection **112** disposed within the distal region **34** occupies within the range of 10%-70% of a volume V.sub.20 of the distal region **34** of the tube assembly **20**, and more particularly within the range of 20%-60% of the volume V.sub.20 of the distal region **34** (see FIG. **5**).

[0042] In one exemplary operation of a conventional ENT shaver, the material is able to penetrate the cutting window to contact the inner tube opposite the cutting window such that the size of the reduced material is approximately equal to the diameter of the lumen. The reduced material having a size approximately equal to the diameter of the lumen increases the likelihood of the reduced material clogging within the lumen, particularly near the cutting window. Furthermore, in instances where the axial length of the cutting window is greater than the diameter of the lumen, the likelihood of the reduced material clogging is further increased in conventional ENT shavers.

[0043] The clog-reducing tip **104** of the present disclosure significantly reduces the likelihood of clogging by, for example, providing that the distance from the proximal boundary **101** of the cutting window **22** at the outer tube **26** to a nearest point on the projection **112** (approximated as point **105** as shown in FIG. **4**) is less than the diameter of the lumen **30**. Thus, any reduced material **106** that may pass through the “throat” (i.e., the distance from cutting window **22** to the nearest point **105**) has a size smaller than the diameter of the lumen **30** itself. Consequently, once the reduced material reaches the lumen **30** proximal to the projection **112**, it is increasingly unlikely that the reduced material clogs the tube assembly **20**.

[0044] Prior to the material being sufficiently reduced to pass through the “throat” of the inner tube **28**, the projection **112** (and the proximal boundary **101** of the cutting window **22**) maintains the material **106** in a position such that the cutting action continues to reduce the material **106** with each rotation of the inner tube **28**. Despite the material **106** possibly remaining positioned near the cutting window **22** for increased time, empirical investigations have shown minimal effect on material removal capacity of the tube assembly **20** incorporating the clog-reducing tip **104** with near or total elimination of clogging commonly associated with conventional ENT shavers.

[0045] With continued reference to FIG. **4** and concurrent reference to FIG. **6**, the projection **112** may extend within the lumen **30** from near the distal end **23** of the tube assembly **20** to a position proximal to (i.e., in the direction of arrow **100**) the proximal boundary **101**. In other words, another portion **113** of the projection **112** may be positioned proximal to the proximal boundary **101**. In other words, the axial length L.sub.112 of the projection **112** may be greater than the axial length L.sub.22 of the cutting window **22**. Positioning the portion **113** of the projection **112** proximal to the proximal boundary **101** ensures that the material **106** passing through the proximal boundary **101** is reduced to a size less than the cross sectional area of the lumen **30** of the tube assembly **20**. It is understood that the axial length L.sub.112 of the projection **112** may be greater than an axial length L.sub.42 of the outer cutting window **42** and/or less than an axial length L.sub.36 of the inner cutting window **36**. In certain embodiments, the projection **112** may extend even more proximally to the proximal boundary **101** than shown in FIGS. **4** and **6** with a proximal second portion **110***b* to be described having a shallower taper.

[0046] The projection **112** of the clog-reducing tip **104** has a shelf or an inner surface **110**. The inner surface **110** is displaced radially inward relative to an interior surface **108** of the lumen **30** (i.e., proximal to the projection **112**) towards the longitudinal axis **24** of the tube assembly **20**. Referring to FIG. **5**, the projection **112** may be angled relative to the interior surface **108** of the

lumen **30**. For example, a line extending between distal and proximal ends of the projection **112** may be oriented at an angle α in the range of approximately 5 degrees to approximately 40 degrees relative to the interior surface **108** of the lumen **30**. In other embodiments, the angle α is between approximately 10 degrees and approximately 30 degrees, and more particularly between approximately 15 degrees and approximately 25 degrees. The angle α generally provides a profile (when viewed in elevation as shown in FIGS. **4-6**) to the projection **112** that tapers in the direction of arrow **100**. In other words, a distance from the longitudinal axis **24** to the projection **112** at the distal boundary **103** of the cutting window **22** is less than a distance from the longitudinal axis **24** from the projection **112** at the proximal boundary **101** of the cutting window **22** such that the projection **112** tapers in the direction towards the proximal boundary **101**. The tapering of the projection **112** advantageously maintains the material **106** near the distal boundary **103** closer to the cutting edges **90** and **94** to reduces the material **106** into smaller bits as the reduced material **106** moves along the inner surface **110** of the projection **112** towards the lumen **30** proximal to the cutting window **22**. The tapering of the projection **112** also ensures a gradual transition through the "throat," as previously described, such that reduced material **106** passing through the "throat" immediately encounters a greater cross sectional area of the lumen **30** and is quickly urged proximally within the lumen **30** under forces from the suction source.
[0047] In certain embodiments, the inner surface **110** of the projection **112** further comprises or is defined by a distal first portion **110***a* and the proximal second portion **110***b* proximal to the distal first portion **110***a*. Referring to FIGS. **5** and **6**. The distal first portion **110***a* may be oriented substantially parallel to the longitudinal axis **24** of the tube assembly **20**. The distal first portion **110***a* may be substantially planar when viewed in elevation. The proximal second portion **110***b* may be sloped or angled relative to the distal first portion **110***a* and the interior surface **108** of the lumen **30**. The proximal second portion **110***b* may be arcuate when viewed in elevation to provide a smooth transition to the distal first portion **110***a*. The proximal second portion **110***b* may be oriented at an angle β in the range of approximately 20 degrees to approximately 60 degrees relative to the interior surface **108** of the lumen **30**. In other embodiments, the angle β is between approximately 30 degrees and approximately 50 degrees.
[0048] Referring to FIG. **7**, the projection **112** may be positioned about the longitudinal axis **24** radially opposite the cutting window **22** of the tube assembly **20**, and more particularly radially opposite the inner cutting window **36**. With the projection **112** within the lumen **30** of the inner tube **28**, the relative positioning between the projection **112** and the inner cutting window **36** remains constant as the inner tube **28** rotates within the outer tube **26**. The radial position of the cutting window **22** of FIG. **7** is approximated between lines W**1** and W**2** with the projection **112** being positioned about the longitudinal axis **24** substantially opposite the space between lines W**1** and W**2**.
[0049] The projection **112** is positioned about a circumference of the lumen **30** in a manner sufficient to suitably reduce the material **106** to prevent clogging of the tube assembly **20**. In certain embodiments, the projection **112** is positioned about less than one half of the circumference of the lumen **30**. For example, the axial cross sectional view FIG. **7** shows one side of the projection **112** radially positioned approximately at the 4 o'clock position, and another side of the projection **112** radially positioned approximately at the 8 o'clock position. Stated differently, an angle γ about the longitudinal axis **24** and extending between opposing sides of the projection **112** within the range of approximately 70 degrees to approximately 180 degrees, and more particularly with the range of approximately 90 degrees to approximately 160 degrees, and even more particularly with the range of approximately 110 degrees to approximately 150 degrees. Other suitable values for the angle γ are contemplated based on, at least in part, the diameter of the lumen **30**, the intended application of the surgical instrument **10**, and the like.
[0050] In certain embodiments, the clog-reducing tip **104** includes an insert secured within the lumen **30** of the inner tube **28**. The insert defines the projection **112** and forms the inner surface

110. For example, the insert may be bonded to the lumen **30** of the inner tube **28**. The insert may include an outer surface **111** and the inner surface **110** with the outer surface **111** shaped to conform a portion of the lumen **30** (see FIG. **7**). The inner surface **110** may define the projection **112**. The insert may have a thickness defined between the inner surface **110** and the outer surface **111** with the thickness of the insert tapering in an axial direction; i.e., the direction **100** towards the proximal boundary **101** of the cutting window **22**. It is also understood that the thickness of the insert may taper radially about the longitudinal axis **24** of the tube assembly **20**, as shown in FIG. **7**.
[0051] In another exemplary embodiment of the clog-reducing tip **104** illustrated in FIGS. **10** and **11**, the projection **112** is defined by the lumen **30** distal to the proximal boundary **101** being formed radially inwardly towards the longitudinal axis **24**. In other words, whereas the projection **112** of FIGS. **4-6** are within the lumen **30** with the inner tube **28** having a generally cylindrical outer profile to the distal end **23** of the tube assembly **20**, the FIGS. **10** and **11** show the inner tube **28** near the distal end **23** (e.g., the distal region **34**) deformed inwardly towards the longitudinal axis **24**. With a portion of the inner tube **28** deformed inwardly, the lumen **30** of the inner tube **28** is corresponding deformed inwardly and consequently defines the projection **112** providing the reduced cross sectional area of the lumen **30** as previously described. The inwardly deformed portion of the inner tube **28** may be constructed through stamping, drawing, or similarly suitable manufacturing process.
[0052] Referring to FIGS. **12-15**, yet another embodiment, according to the present disclosure, of the clog-reducing tip **104** is shown. In this embodiment, the clog-reducing tip **104** includes the inner surface **110** formed by boring out an eccentric borehole within the distal region **34** of the inner tube **28**. The borehole is eccentric with respect to the longitudinal axis **24** of the tube assembly **20**. The eccentric borehole is in communication with the cutting window **22** and the lumen **30** of the inner tube **28**. This type of tip would typically be machined. As illustrated, the inner surface **110** has a smaller cross-section at the first cutting window **36**. The process of machining the clog-reducing tip **104** is illustrated in FIGS. **16-19**.
[0053] The present disclosure provides a method, according to one embodiment of the present disclosure, for operating the surgical instrument **10** on a patient. The method includes the steps of providing the cutting assembly **14** including the tube assembly **20** extending axially. The tube assembly **20** includes the rotatable inner tube **28** having the lumen **30** disposed coaxially within the outer tube **26**. The inner tube **28** forms the inner cutting window **36**, and the outer tube **26** forms the outer cutting window **42**. The inner and outer cutting windows **36**, **42** define the cutting window **22** of the tube assembly **20**. The method may also includes the steps of providing the projection **112** within the lumen **30** of the inner tube **28** with at least a portion of the projection **112** disposed within the distal region **34** of the inner tube **28** with the projection **112**. In certain embodiments, the projection **112** positioned distal to the proximal boundary **101** of the cutting window **22**. The projection **112** provides a reduced cross sectional area to the inner tube **28** relative to a cross sectional area of the lumen proximal to the projection **112**. In certain embodiments, the projection **112** is the volume V.sub.112 that occupies the volume V.sub.20 of the lumen **30** distal to the proximal boundary **101** of the cutting window **22**. The method includes the step of applying the cutting window to a surgical site of a patient and rotating the inner tube **28** relative to the outer tube **26** by the drive assembly **12** to cut the material **106** by an interaction of the inner cutting window **36** and the outer cutting window **42** on the patient, wherein the projection **112** reduces the size of the material **106** removed through the cutting window **22** to reduce clogging of the tube assembly **20**.
[0054] The surgical instrument **10** of the present disclosure also advantageously reduces the likelihood of clogging at or just proximal to the tube assembly **20** of the surgical instrument **10**. FIG. **20** is a cross sectional view of a portion of the surgical instrument of FIG. **1**, showing in particular an interface **114** between the tube assembly **20** and the drive hub **48**. As mentioned, the hub member **50** of the drive hub **48** includes the reduced diameter portion **56** defining the reduced

aperture **53** in communication with the aperture **52** of the drive hub **48** (and the connecting hub **68**). The lip **55** is formed by the decrease in diameter from the aperture **52** to the reduced aperture **53**. The inner tube **28** has an axial length longer than an axial length of the outer tube **26** such that the inner tube **28** extends past the proximal region **38** of the outer tube **26** and into the connecting hub **68** and the drive hub **48**, as shown in FIG. **20**.

[0055] During assembly of the surgical instrument **10**, such as when coupling the tube assembly **20** with the drive hub **48**, the inner tube **28** is slidably inserted within the aperture **52** of the drive hub **48** and positioned adjacent or in an abutting relationship with the lip **55**. The lip **55** facilitates appropriate axial positioning the tube assembly **20** relative to the housing **15** and other structures of the surgical instrument **10**. The lumen **30** of the inner tube **28** is in fluid communication with the reduced aperture **53** of the drive hub **48**, as shown in FIG. **20**, such that reduced material **106** may pass from the lumen **30** to the suction source.

[0056] The diameter of the lumen **30** of the inner tube **28** is less than the diameter of the reduced aperture **53** at the interface **114**. In other words, the reduced material **106** moves from a smaller cross sectional area of the lumen **30** to a greater cross sectional area of the reduced aperture **53** as the material **106** passes through the interface **114**. In effect, the passage through which the reduced material is moving expands, thereby reducing the likelihood of clogging. If, for a contrasting example, the diameter of the lumen **30** of the inner tube **28** was greater than the diameter of the reduced aperture **53**, the reduced material **106** may become lodged on the lip **55** and increase the likelihood of clogging at the interface **114**.

[0057] Thus, according to one exemplary embodiment of the present disclosure, a cutting assembly for a surgical instrument having a drive assembly, said cutting assembly comprising: a tube assembly comprising a cutting window near a distal end and adapted to be applied to a surgical site of a patient, an outer tube, an inner tube coaxially disposed within and rotatable relative to said outer tube with the drive assembly with said inner tube comprising a lumen; and a drive hub coupled to said inner tube with said drive hub defining an aperture adapted to slidably receive a proximal end of said inner tube, and defining a reduced aperture in communication with said aperture, wherein a diameter of said reduced aperture is less than a diameter of said aperture, wherein a diameter of said lumen is less than said diameter of said reduced aperture when said proximal end of said inner tube is slidably received within said aperture to reduce clogging of said surgical instrument as removed material moves from said lumen to said reduced aperture of said drive hub. A lip is formed at an interface between said aperture and said reduced aperture with said proximal end of said inner tube adapted to be positioned adjacent to said lip.

[0058] Accordingly, the surgical instrument **10** of the present disclosure reduces the occurrence of the clogging by providing the clog-reducing tip **104** having the projection **112** for reducing a cross sectional area of the lumen **30** distal to the proximal boundary **101** of the cutting window **22** and/or for providing the volume V.sub.112 within the volume V.sub.20 of the distal region **34** of the tube assembly **20**. The size of the material **106** that may enter the distal region **34** of the inner tube **28** is limited and maintained in a position to be further reduced by the cutting action. Further, only material **106** of sufficiently reduced sized may pass through the “throat” of the tube assembly **20**, after which the reduced material **106** encounters the larger cross sectional of the lumen **30** also under the influence of suction. The projection **112** may be the insert secured with the lumen **30** of the inner tube **28**, or formed integrally with the same, such as by deforming the distal region **34** of the inner tube **28**, providing the borehole eccentric to the longitudinal axis **24** of the tube assembly **20**, or suitably milling within the inner tube **28** to define the projection **112**. The surgical instrument **10** of the present disclosure cuts and aspirates tissue as per current shaver systems utilizing suction. It should be appreciated that, in another embodiment, the surgical instrument **10** may be used with the surgical tools or be a dedicated tool or instrument.

[0059] It will be further appreciated that the terms “include,” “includes,” and “including” have the same meaning as the terms “comprise,” “comprises,” and “comprising.”

[0060] The present invention has been described in an illustrative manner. It is to be understood that the terminology, which has been used, is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present invention are possible in light of the above teachings. Therefore, the present invention may be practiced other than as specifically described.
[0061] Embodiments of the disclosure may be described with reference to the following exemplary clauses:
[0062] Clause 1—A cutting assembly for a surgical instrument for cutting tissue, said cutting assembly being configured to be coupled to a drive assembly including a motor having a rotatable drive element enclosed in a housing and said cutting assembly comprising: a rotatable first tube having a lumen, said lumen having a proximal region and a distal region, said first tube forming a first cutting window in said distal region; a second tube disposed over said first tube, said second tube having a proximal region and a distal region, said second tube forming a second cutting window in said distal region; said first tube being rotatable relative to said second tube; said proximal region of said window of said lumen having a cross-sectional area greater than a cross-sectional area of said distal region of said lumen such that tissue cut by an interaction of said first cutting window and said second cutting window is of suitable dimensions to allow passage through said first cutting window and said distal region of said lumen to said proximal region of said lumen to prevent clogging of said distal region of said lumen.
[0063] Clause 2—A cutting assembly as set forth in clause 1 wherein said proximal region of said lumen has an interior surface and said distal region of said lumen has an inner surface opposite said first cutting window.
[0064] Clause 3—A cutting assembly as set forth in clause 2 wherein said inner surface is displaced radially inward relative to said interior surface.
[0065] Clause 4—A cutting assembly as set forth in clause 2 wherein said inner surface extends radially and axially at an angle greater than zero relative to said interior surface.
[0066] Clause 5—A cutting assembly as set forth in clause 2 including an insert disposed within said distal region of said lumen opposite said first cutting window and forming said inner surface.
[0067] Clause 6—A cutting assembly as set forth in clause 5 wherein said insert is bonded to said first tube.
[0068] Clause 7—A cutting assembly as set forth in clause 5 wherein said insert includes said inner surface extending radially and axially at an angle greater than zero relative to said interior surface.
[0069] Clause 8—A cutting assembly as set forth in clause 5 wherein said insert has one of a generally arcuate, semi-circular, and rectangular cross-sectional profile.
[0070] Clause 9—A cutting assembly as set forth in clause 5 wherein said insert is made of one or more different materials.
[0071] Clause 10—A cutting assembly as set forth in clause 2 wherein said inner surface is defined by said inner tube in said distal region of said lumen opposite said first cutting window.
[0072] Clause 11—A cutting assembly as set forth in clause 2 wherein said inner surface extends axially from a distal end of said distal region of said lumen to one of less than and at least a proximal end of said first cutting window.
[0073] Clause 12—A cutting assembly as set forth in clause 11 wherein said first cutting window has an axial length less than an axial length of one of said inner surface and said second cutting window.
[0074] Clause 13—A cutting assembly as set forth in clause 11 wherein an angle of said inner surface to a tube wall of said distal region of said lumen is between approximately 20 degrees and approximately 90 degrees.
[0075] Clause 14—A cutting assembly as set forth in clause 11 wherein said inner surface has a radial height greater than a radial height of said interior surface.
[0076] Clause 15—A cutting assembly as set forth in clause 1 including a third tube disposed over

said second tube.

[0077] Clause 16—A cutting assembly as set forth in clause 1 wherein said distal region of said lumen has a profile formed by one of a drawing process and a machined process.

[0078] Clause 17—A cutting assembly as set forth in clause 16 wherein said distal region of said lumen has a non-circular cross-section.

[0079] Clause 18—A cutting assembly as set forth in clause 1 wherein said first cutting window includes at least one cutting edge.

[0080] Clause 19—A cutting assembly as set forth in clause 1 including an aspiration path connected to either one of said first tube and said second tube.

[0081] Clause 20—A cutting assembly as set forth in clause 1 wherein a cross-section of said distal region of said lumen and a cross-section of said proximal region of said lumen has a ratio of one of 1:1.5, 1:3, and 1:6.

[0082] Clause 21—A cutting assembly as set forth in clause 1 wherein an axial length of said first cutting window relative to a diameter of said distal region of said lumen is such that no dimension of a bone chip that is cut is larger than a diameter of said lumen in said proximal region.

[0083] Clause 22—A surgical instrument for use on a patient, said surgical instrument comprising: a cutting assembly including a plurality of tubes extending axially, said tubes comprising at least a rotatable inner tube having a lumen, said lumen having a proximal region and a distal region, said inner tube forming an inner cutting window in said distal region, an outer tube disposed over said inner tube, said outer tube having a proximal region and a distal region, said outer tube forming an outer cutting window in said distal region, said inner tube being rotatable relative to said outer tube; a drive assembly including a motor having a rotatable drive element, a housing for enclosing said motor and being removably coupled to said cutting assembly, a suction connection on said housing for connection to a suction source, and a suction passage extending from said inner window through said inner tube and through said housing to said suction connection; an irrigation connection on said housing for connection to a fluid source; an irrigation passage extending through said housing between said irrigation connection and said cutting assembly and between said inner tube and said outer tube to said cutting window to provide lubrication and flush blood, tissue, and bone; a suction connection on said housing for connection to a suction source; a suction passage extending through said housing between said suction connection and said cutting window of said inner tube; and said proximal region of said lumen having a cross-sectional area greater than a cross-sectional area of said distal region of said lumen such that tissue cut by an interaction of said inner cutting window and said outer cutting window of suitable dimensions to allow passage through said inner cutting window and said distal region of said lumen to said proximal region of said lumen to prevent clogging of said distal region of said lumen.

[0084] Clause 23—A method of operating a surgical instrument for use on a patient, said method comprising the steps of: providing a cutting assembly including a plurality of tubes extending axially, the tubes comprising at least a rotatable inner tube having a lumen, the lumen having a proximal region and a distal region, the inner tube forming an inner cutting window in the distal region, an outer tube disposed over the inner tube, the outer tube having a proximal region and a distal region, the outer tube forming an outer cutting window in said distal region, the inner tube being rotatable relative to the outer tube; providing a drive assembly including a motor having a rotatable drive element, a housing for enclosing the motor and being removably coupled to the cutting assembly, a suction connection on the housing for connection to a suction source, and a suction passage extending from the inner window through the inner tube and through the housing to the suction connection; providing the proximal region of the lumen having a cross-sectional area greater than a cross-sectional area of the distal region of the lumen; rotating the inner tube relative to the outer tube by the drive assembly; cutting bone and/or tissue by an interaction of the inner cutting window and the outer cutting window on the patient; and allowing passage of cut bone and/or tissue of suitable dimensions through the inner cutting window and the distal region of the

lumen to the proximal region of the lumen to prevent clogging of the distal region of the lumen.
[0085] Clause 24—A surgical instrument, cutting assembly, and method as disclosed and described herein, including equivalents not specifically recited herein.

Claims

1. A cutting assembly configured to be removably coupled to a drive assembly of a surgical instrument, the cutting assembly comprising: a connecting hub; a tube assembly extending distally from the connecting hub and defining a cutting window, wherein the tube assembly comprises an outer tube, and an inner tube coaxially disposed within the outer tube and defining a lumen in communication with the cutting window, wherein the inner tube is configured to be rotatable relative to the outer tube about a longitudinal axis; and a clog-reducing tip comprising a projection disposed within the inner tube, wherein at least a portion of the projection is positioned distal to a proximal boundary of the cutting window.
2. The cutting assembly of claim 1, wherein the projection is positioned about the longitudinal axis radially opposite the cutting window of the tube assembly.
3. The cutting assembly of claim 1, wherein a distance between the proximal boundary of the cutting window and a nearest point on the projection is less than a diameter of the lumen.
4. The cutting assembly of claim 1, wherein the projection is an insert rigidly coupled to the inner tube.
5. The cutting assembly of claim 4, wherein the insert comprises an outer surface conforming and bonded to an interior surface of the inner tube.
6. The cutting assembly of claim 1, wherein the projection is formed by the inner tube being deformed radially inwardly towards the longitudinal axis.
7. The cutting assembly of claim 1, wherein an axial length of the cutting window is greater than a diameter of the lumen.
8. The cutting assembly of claim 1, wherein an axial length of the projection is greater than an axial length of the cutting window.
9. The cutting assembly of claim 1, wherein the projection comprises a distal first portion, and a proximal second portion that is angled relative to the distal first portion.
10. The cutting assembly of claim 9, wherein the proximal second portion is arcuate to provide a smooth transition to the distal first portion.
11. The cutting assembly of claim 9, wherein at least a portion of the proximal second portion is positioned proximal to the proximal boundary.
12. The cutting assembly of claim 1, wherein a distal region of the tube assembly is defined distal to the proximal boundary, and wherein the projection occupies a volume of the distal region within a range of approximately 10%-70%.
13. A cutting assembly configured to be removably coupled to a drive assembly of a surgical instrument, the cutting assembly comprising: a connecting hub; a tube assembly extending distally from the connecting hub and defining a cutting window, wherein the tube assembly comprises an outer tube, and an inner tube coaxially disposed within the outer tube and defining a lumen in communication with the cutting window, wherein the inner tube is configured to be rotatable relative to the outer tube about a longitudinal axis; and an insert bonded to an inner surface of the inner tube and at least partially disposed axially within the cutting window.
14. The cutting assembly of claim 13, wherein the insert is bonded to a portion of the inner surface that is radially opposite the cutting window.
15. The cutting assembly of claim 13, wherein the insert comprises a distal first portion, and a proximal second portion that is angled relative to the distal first portion.
16. The cutting assembly of claim 15, wherein the proximal second portion tapers in a proximal direction.

17. The cutting assembly of claim 13, wherein an axial length of the insert is greater than an axial length of the cutting window.

18. A cutting assembly configured to be removably coupled to a drive assembly of a surgical instrument, the cutting assembly comprising: a connecting hub; a tube assembly extending distally from the connecting hub and defining a cutting window, wherein the tube assembly comprises an outer tube, and an inner tube coaxially disposed within the outer tube and defining a lumen in communication with the cutting window, wherein the inner tube is configured to be rotatable relative to the outer tube about a longitudinal axis; and a projection disposed within the inner tube and defining a throat having a distance between a proximal boundary of the cutting window on the outer tube and a nearest point on the projection, wherein the throat is narrower than a diameter of the lumen.

19. The cutting assembly of claim 18, wherein the projection comprises a distal first portion, and a proximal second portion that is angled relative to the distal first portion, wherein the proximal second portion comprises the nearest point.

20. The cutting assembly of claim 19, wherein the distal first portion is oriented substantially parallel to the longitudinal axis so as to maintain material in position for cutting action to continue until the material is sufficiently reduced to pass through the throat.